

Unit 109 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 72.9 ac. 34 plots, BAF 20, variable-radius cruise on a 300 ft grid, 364 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (1 SE of mean, BA 4.3 %)	Type	Sierran mixed conifer
Trees per acre	65	CWHR size / density	5 / D (cover 88 %)
QMD	24.6 in	SDI	263 (44 % of max 592)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	213 sq ft/ac 25.1 in	BA QMD	1 sq ft/ac 8.8 in
TPA	62	TPA	3
Volume	38.1 MBF/ac (7598 cu ft)	Volume	0.6 green tons/ac (28 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 169 sq ft/ac BA	Vol. sampling error	9.7 % at 95 % (CV 28 %, t = 2.03, n = 34)
Species (BA)	WF 70, PP 55, DF 36, IC 32, SP 21	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	5.3 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	8.8 % at 95 % (CV 25 %); reference, not the standard
		Measured trees	PP 94, WF 119, DF 62, SP 35, IC 54; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	34; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	4.3	2.4	61	0.3
12-15.9	17	9.4	10.0	233	1.2
16-19.9	40	12.7	23.5	661	3.3
20-23.9	68	15.6	40.0	1236	6.2
24-27.9	44	7.1	25.9	830	4.2
28-31.9	49	6.0	28.8	1076	5.4
32-35.9	39	3.7	22.9	875	4.4
36-39.9	31	2.3	18.2	748	3.8
40-43.9	25	1.5	14.7	631	3.2
44-47.9	13	0.7	7.6	325	1.6
48-51.9	9	0.4	5.3	234	1.2
52-55.9	9	0.3	5.3	250	1.3
56-59.9	6	0.2	3.5	169	0.9
60-63.9	10	0.3	5.9	298	1.5
All	364	64.7	214.1	7627	38.3

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	119	21.1	70.0	24.7	2480	12.5
PP	94	16.0	55.3	25.2	2184	11.0
DF	62	11.2	36.5	24.4	1245	6.3
IC	54	10.5	31.8	23.5	929	4.7
SP	35	5.9	20.6	25.3	789	4.0
All	364	64.7	214.1	24.6	7627	38.3

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 109 plots, 300 ft grid (orange: QA flag)

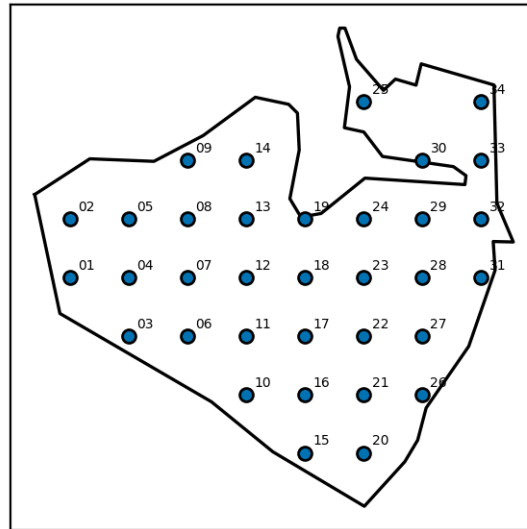


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
109-01	15	300	8	ok
109-02	10	200	7	ok
109-03	10	200	21	ok
109-04	11	220	7	ok
109-05	9	180	7	ok
109-06	12	240	19	ok
109-07	3	60	15	ok
109-08	12	240	4	ok
109-09	12	240	17	ok
109-10	11	220	9	ok
109-11	13	260	16	ok
109-12	11	220	6	ok
109-13	15	300	11	ok
109-14	13	260	3	ok
109-15	10	200	17	ok
109-16	10	200	7	ok
109-17	10	200	11	ok
109-18	10	200	7	ok
109-19	7	140	35	ok
109-20	12	240	16	ok
109-21	12	240	8	ok
109-22	9	180	7	ok
109-23	11	220	5	ok
109-24	13	260	24	ok
109-25	10	200	24	ok
109-26	10	200	21	ok
109-27	10	200	11	ok
109-28	14	280	12	ok
109-29	11	220	21	ok
109-30	11	220	35	ok
109-31	13	260	25	ok
109-32	2	40	21	ok
109-33	12	240	34	ok
109-34	10	200	14	ok